CAPSTONE PROJECT TECHNICAL INFORMATION REPORT

# Detecting Depression Using Multimodal Vocal, Acoustic & Textual Cues

**Course:** UE23CS320B — Capstone Project | **Project ID:** PW26_CBH_04 | **Guide:** Ms. Chandravva Hebbi

## Part 1 — Project Title & Basic Information

| Parameter | Verified Project Metadata |
|---|---|
| **Exact Project Title** | Detecting Depression using vocal cues |
| **Project ID** | PW26_CBH_04 |
| **Course / Subject** | UE23CS320B – Capstone Project (Phase 2 & Phase 3) |
| **Project Guide** | Ms. Chandravva Hebbi |
| **Team Members** | Vishruth Rallapalli, Yuvraj Rao, Vivan Jeevan Naik, Srujan Kumar J |
| **Target Dataset** | DAIC-WOZ (Distress Analysis Interview Corpus — Wizard of Oz) |
| **Clinical Metric Target** | PHQ-8 (Patient Health Questionnaire-8) 3-Class Severity, Binary Status, & Continuous Score |

## Part 2 — Abstract & Technical Scope

**Abstract:** This project presents a participant-independent, multimodal machine learning screening prototype for depression detection and severity assessment using the DAIC-WOZ dataset. The system processes semi-structured interview audio recordings (.wav) and matching transcripts (.csv). The pipeline extracts three distinct information streams: (1) 154-dimensional acoustic features comprising 88 openSMILE eGeMAPS features, 52 MFCC statistics, and 14 prosodic/spectral statistics; (2) 768-dimensional contextual speech representations from frozen Wav2Vec 2.0 (`facebook/wav2vec2-base-960h`); and (3) 768-dimensional contextual linguistic representations extracted from participant-only speech using frozen BERT (`bert-base-uncased`). The standardized 1,690-dimensional multimodal embedding feeds into the final **CompactTriModalMLP** (225,413 parameters) that simultaneously predicts: (a) 3-class categorical depression severity (Minimal/Mild, Moderate, Moderately Severe/Severe), (b) binary clinical depression status (PHQ-8 ≥ 10), and (c) continuous PHQ-8 score regression (0–24 scale).

**Scope & Limitations:** The system is strictly intended as a preliminary depression screening / research decision-support prototype to assist clinicians by extracting objective paralinguistic and semantic cues. It is *not* an autonomous diagnostic authority. The prototype operates within the empirical constraints of the DAIC-WOZ dataset (183 development sessions, 9 reserved unseen test sessions) and incorporates training-set class weighting to handle an 84.15% majority class skew.

## Part 3 — Previous Work vs. Current Implementation

**Previously Implemented (Phase 1 & Phase 2 Baseline):** Initial work investigated single-modality acoustic pipelines (basic MFCC extraction via Librosa, pitch tracking) and an 847,749-parameter recurrent BiLSTM with modality gating. Experimental evaluation revealed that the 847k-parameter BiLSTM severely overfit the small training cohort (116 sessions), collapsing into the 84% majority Minimal/Mild class (achieving 83.78% accuracy but 0.00% recall on moderate and severe depression on holdout data).

**Currently Implemented in Final Model:**

- **Interviewer Speech Filtering:** Automated transcript parsing extracts strictly participant utterances (`speaker == 'Participant'`), discarding Ellie interviewer speech to eliminate dialogue-structure shortcut learning.
- **Tri-Modal Feature Standardization:** Standardized fusion of 154d Acoustic (88 eGeMAPS + 52 MFCC stats + 14 prosodic stats) + 768d Wav2Vec 2.0 + 768d BERT embeddings (1,690 dimensions).
- **Compact MLP Replacement:** Replaced the overparameterized 847k-parameter BiLSTM with the 225,413-parameter **CompactTriModalMLP** regularized with LayerNorm and Dropout.
- **Leak-Free 5-Fold Stratification:** Implemented a 5-fold Stratified Group Split ensuring strict participant isolation and valid minority representation across folds.

- **Macro-F1 Model Selection:** Checkpoints are selected based on Validation Macro-F1 and Balanced Accuracy rather than raw cross-entropy loss.

## Part 4 — Literature Survey & Inferences

| Reference | Key Methodology | Inference & Impact on Project |
|---|---|---|
| **Alghowinem et al. (2013) IEEE ICASSP** | Analyzed acoustic differences between spontaneous and read clinical speech. | Observed flattened fundamental frequency (F0) and reduced formant dynamics in depressed speech. Supported extracting pitch variance and openSMILE acoustic functionals. |
| **Cummins et al. (2015) Speech Communication** | Reviewed depression and suicide risk assessment using speech analysis. | Identified vocal tract psychomotor retardation (slower speech rate, extended pause duration, spectral energy drop) as valid clinical indicators. |
| **Morales et al. (2018) Interspeech / ACM MM** | Multimodal fusion of acoustic features and linguistic transcripts. | Demonstrated that combining textual semantics with speech acoustics outperforms unimodal acoustic baselines, informing our BERT + Wav2Vec2 fusion design. |
| **Ringeval et al. (2019) AVEC / ACM MM** | AVEC Depression Sub-challenge benchmark on DAIC-WOZ. | Provided the baseline eGeMAPS acoustic feature protocol and PHQ-8 clinical severity categorization standards (0-9, 10-14, 15-24). |

## Part 5 — Complete System Architecture & Data Flow

The final verified architecture follows a leak-free multimodal pipeline:

| Pipeline Stage | Component / Technology | Input → Output | Verified Details |
|---|---|---|---|
| **1. Raw Ingestion** | DAIC-WOZ Reader (Librosa, Pandas) | Raw .wav + .csv → Audio array & transcript dataframe | Session validation against master index; 9 reserved unseen PIDs isolated. |
| **2. Preprocessing** | Audio Cleaner & Text Filter (AudioCleaner, BertExtractor) | Raw audio/text → Participant-only speech & text | 16 kHz resampling, silence trimming, Ellie speech discarded (`speaker == 'Participant'`). |
| **3. Feature Extraction** | eGeMAPS + Wav2Vec2 + BERT Extractor | Audio/Text → 154d Acoustic, 768d W2V2, 768d BERT | 154d Acoustic (88 eGeMAPS + 52 MFCC + 14 prosodic stats); frozen Wav2Vec 2.0; frozen BERT-base. |
| **4. Scaling & Fusion** | StandardScaler + Concatenation | 3 feature sets → 1,690d Standardized Fused Vector | `StandardScaler` fit strictly on `train_df` (1690d); transformed on val/test/unseen. |
| **5. Final Model** | CompactTriModalMLP (PyTorch) | 1,690d Vector → 3-Class Softmax, Binary Prob, PHQ-8 Score | Linear(1690→128) → LayerNorm → ReLU → Dropout(0.3) → Linear(128→64) → LayerNorm → 3 Heads. |

## Part 6 — Dataset Statistics & Partitions

| Partition | Sessions | Minimal / Mild (0-9) | Moderate (10-14) | Mod.Severe / Severe (15-24) | Gender (M / F) |
|---|---|---|---|---|---|
| **Development Master Index** | 183 (100.0%) | 154 (84.15%) | 20 (10.93%) | 9 (4.92%) | 113 M / 70 F |
| **Training Split** | 116 (63.4%) | 98 (84.5%) | 12 (10.3%) | 6 (5.2%) | 72 M / 44 F |
| **Validation Split** | 30 (16.4%) | 25 (83.3%) | 4 (13.3%) | 1 (3.3%) | 18 M / 12 F |
| **Test Holdout Split** | 37 (20.2%) | 31 (83.8%) | 4 (10.8%) | 2 (5.4%) | 23 M / 14 F |
| **Reserved Unseen Set** | 9 (Isolated) | 4 (44.4%) | 2 (22.2%) | 3 (33.3%) | 5 M / 4 F |

## Parts 7, 8, 9 & 10 — Modality Extraction Pipelines & Acoustic Inventory

- **Audio Pipeline (Wav2Vec 2.0):** Ingests 16 kHz resampled audio. Frozen `facebook/wav2vec2-base-960h` extracts contextual speech representations pooled into a 768-dimensional vector per session.

• **Transcript Pipeline (BERT):** Participant utterances are extracted by filtering `speaker == 'Participant'`. Frozen `bert-base-uncased` processes text in 512-token sliding windows with 128-token overlap, mean-pooling `[CLS]` tokens into a 768-dimensional contextual linguistic vector per session.

• **154-Dimensional Acoustic Feature Breakdown:**

| Feature Group | Dimensions | Detailed Processing | Clinical Indicator |
|---|---|---|---|
| **eGeMAPS v02 Set** | 88 features | openSMILE standard set: Pitch (F0), jitter, shimmer, loudness, alpha ratio, Hammarberg index, formants F1-F3. | Physiological biomarkers of psychomotor vocal tract slowing and vocal fold tension. |
| **MFCC Statistics** | 52 features (26 mean + 26 std) | Librosa frame extraction (25ms window, 10ms hop): 13 static MFCCs + 13 delta MFCCs (mean & std across frames). | Captures articulation dynamics and spectral timbre variability. |
| **Prosodic & Spectral Statistics** | 14 features (7 mean + 7 std) | RMS Energy, F0 fundamental frequency, Spectral Centroid, Spectral Rolloff, Zero Crossing Rate (mean & std across frames). | Monitors flat affect, low acoustic energy, and reduced pitch modulation. |

## Parts 11, 12 & 13 — Model Inventory & Final CompactTriModalMLP Configuration

**Model Parameter Comparison:** Earlier BiLSTM baseline = **847,749 parameters** (replaced due to severe overfitting). Final **CompactTriModalMLP = 225,413 total parameters** (225,413 trainable | Encoders frozen).

| Configuration Property | Verified Value / Setting | Implementation Details |
|---|---|---|
| **Final Model Name** | **CompactTriModalMLP** | 2-layer regularized multi-task Multimodal MLP. |
| **Input Representation** | 1,690-dim Fused Vector | 154d Acoustic + 768d Wav2Vec2 + 768d BERT (StandardScaled). |
| **Hidden Layers & Regularization** | Linear(1690→128) → LayerNorm → ReLU → Dropout(0.3) → Linear(128→64) → LayerNorm → ReLU | LayerNorm and 30% dropout prevent co-adaptation on small training cohorts. |
| **Loss Function** | Multi-Task: `L = CrossEntropy(weight=W) + 0.5*BCE + 0.2*MSE(y/24)` | Combines 3-class severity classification, binary depression diagnosis, and normalized PHQ regression. |
| **Class Weights (Train Only)** | **Class 0: 0.3946 \| Class 1: 3.2222 \| Class 2: 6.4444** | Inverse-frequency weights computed strictly on `train_df` (98 / 12 / 6 distribution). |
| **Optimizer & Schedule** | AdamW (lr=1e-3, weight_decay=1e-3) + CosineAnnealingLR (T_max=40) | Smooth learning rate decay with batch size = 16. |
| **Checkpoint Selection** | Primary: **Validation Macro-F1** \| Secondary: **Balanced Accuracy** | Prevents majority-class collapse by rejecting models with 0 minority recall. |
| **Post-Hoc Calibration** | Post-hoc Temperature Scaling: **T = 1.6742** | Optimized on validation split logits via L-BFGS. |

## Parts 14, 15, 16 & 17 — Experimental Results & Confusion Matrices

**Final Verified Results Across Splits (CompactTriModalMLP, Seed 42):**

| Metric | Train Split (116) | Validation Split (30) | Test Holdout (37) | Majority Baseline |
|---|---|---|---|---|
| **3-Class Accuracy** | 100.00% | 80.00% | **83.78%** | 83.78% (31/37) |
| **3-Class Macro-F1** | 100.00% | 57.11% | **42.36%** | 30.39% |
| **Balanced Accuracy** | 100.00% | 71.00% | **48.92%** | 33.33% |
| **Binary Accuracy (≥10)** | 100.00% | 76.67% | **86.49%** | 83.78% |
| **Binary F1 (≥10)** | 100.00% | 36.36% | **54.55%** | 0.00% |
| **Severe Class Recall** | 100.00% (6/6) | 100.00% (1/1) | **50.00% (1/2)*** | 0.00% (0/2) |
| **PHQ-8 Score MAE** | 2.91 pts | 3.09 pts | **3.07 pts** | 3.32 pts |

**Note on Severe Recall: 50.00% reflects 1 out of 2 severe test participants correctly classified; small sample denominator noted.*

### Confusion Matrices (Rows = True Class, Columns = Predicted Class):

| Split | True Class \ Predicted Class | Minimal/Mild (0) | Moderate (1) | Mod.Severe/Severe (2) |
|---|---|---|---|---|
| **Train (116)** | Minimal/Mild (0)<br>Moderate (1)<br>Severe (2) | 98<br>0<br>0 | 0<br>12<br>0 | 0<br>0<br>6 |
| **Val (30)** | Minimal/Mild (0)<br>Moderate (1)<br>Severe (2) | 22<br>3<br>0 | 1<br>1<br>0 | 2<br>0<br>1 |
| **Test (37)** | Minimal/Mild (0)<br>Moderate (1)<br>Severe (2) | 30<br>2<br>1 | 0<br>0<br>0 | 1<br>2<br>1 |

## Part 18 — Evaluation on 9 Reserved Unseen Clinical Sessions

The 9 sessions were permanently isolated from training, validation, scaling, threshold tuning, and calibration. The table below presents the verified single-session evaluation results from `evaluate_unseen.py`:

| Session / PID | Words | Actual Severity | Predicted Severity | Outcome | Softmax Conf. | GT / Pred PHQ |
|---|---|---|---|---|---|---|
| **304_AUDIO** | 976 | Minimal / Mild (6) | Minimal / Mild | **CORRECT** | 90.3% | 6 / 3.2 |
| **397_AUDIO** | 1,654 | Minimal / Mild (5) | Minimal / Mild | **CORRECT** | 93.8% | 5 / 2.9 |
| **303_AUDIO** | 1,960 | Minimal / Mild (0) | Minimal / Mild | **CORRECT** | 84.9% | 0 / 3.4 |
| **408_AUDIO** | 849 | Minimal / Mild (1) | Minimal / Mild | **CORRECT** | 93.9% | 1 / 2.8 |
| **409_AUDIO** | 2,352 | Moderate Depression (10) | Moderate Depression | **CORRECT** | 71.6% | 10 / 3.5 |
| **321_AUDIO** | 910 | Mod.Severe / Severe (20) | Mod.Severe / Severe | **CORRECT** | 87.7% | 20 / 3.5 |
| **300_AUDIO** | 352 | Moderate Depression (12) | Minimal / Mild | INCORRECT | 94.8% | 12 / 2.8 |
| **338_AUDIO** | 839 | Mod.Severe / Severe (15) | Minimal / Mild | INCORRECT | 84.1% | 15 / 3.1 |
| **414_AUDIO** | 1,442 | Mod.Severe / Severe (16) | Minimal / Mild | INCORRECT | 94.0% | 16 / 2.6 |

**Overall Unseen Accuracy: 6 / 9 (66.67%)** | Correctly identified 4/4 Minimal/Mild, 1/2 Moderate (PID 409 at 71.6% conf), and 1/3 Severe (PID 321 at 87.7% conf).

## Parts 19, 20, 21 & 22 — Modules, Testing & Technologies

| Module Name | Purpose | Input → Output | Technologies Used |
|---|---|---|---|
| **Feature Store** | Compressed caching of multimodal features. | Raw audio/text → `.npz` feature files | NumPy, Joblib |
| **BERT Extractor** | Participant-only linguistic feature extraction. | .csv transcript → 768d CLS embedding | HuggingFace Transformers, PyTorch |
| **Wav2Vec Extractor** | Self-supervised speech representation extraction. | 16 kHz audio → 768d speech embedding | Torchaudio, PyTorch |
| **eGeMAPS Extractor** | Standard Geneva acoustic biomarker extraction. | 16 kHz audio → 88d acoustic vector | openSMILE, Librosa |
| **Training Engine** | Leak-free training, stratification & benchmarking. | Features → `best_depression_model.pth` | PyTorch, Scikit-learn |
| **Single Evaluator** | Inference on single session recordings. | .wav + .csv → Clinical severity report | PyTorch, Joblib, Librosa |
| **Unseen Benchmark** | Automated evaluation on reserved unseen dataset. | `unseen/` folder → Full clinical report | Pandas, PyTorch |

## Parts 23, 24, 25 & 26 — Team Roles, Demonstrables & Limitations

### Team Task Distribution:

- **Vivan Jeevan Naik:** Multi-modal neural architecture design, Tri-Modal feature fusion, leak-free Stratified Group Splitting, and training pipeline optimization.
- **Vishruth Rallapalli:** Acoustic feature extraction pipeline (eGeMAPS 88d, Librosa temporal features), audio cleaner, and silence suppression.
- **Yuvraj Rao:** Transcript parsing, participant-only text extraction (`speaker == 'Participant'`), and BERT embedding pipeline.
- **Srujan Kumar J:** Unseen dataset benchmarking engine, calibration (TemperatureScaler), and clinical metric evaluation suite.

*Note on Development Work, LOC & Time Spent:* Exact individual hours and lines of code are not formally tracked in the repository; modules were integrated pair-wise.

### Demonstrable CLI Commands:

1. Training Execution: `PYTHONPATH=app python3 app/src/models/train_leak_free.py 40` (Trains CompactTriModalMLP, saves checkpoint, reports Train/Val/Test metrics).
2. Unseen Batch Report: `PYTHONPATH=app python3 app/evaluate_unseen.py --full-report` (Evaluates 9 reserved sessions, outputs 66.67% accuracy).
3. Single-Session Prediction: `PYTHONPATH=app python3 app/predict_single_audio.py unseen/severe/321_AUDIO_interviewee.wav` (Outputs instant clinical prediction).

### Documented Limitations:

- **Small Sample Size & Class Imbalance:** Only 9 severe depression sessions exist in the 183-session development corpus (4.92%), making statistical convergence of deep sequential models challenging.
- **Moderate Class Transition Overlap:** Moderate depression (PHQ-8 10-14) exhibits boundary overlap with both mild and severe classes, resulting in borderline classification confusion.

## Part 27 — References (IEEE Format)

[1] S. Alghowinem, R. Goecke, M. Wagner, G. Parker, and M. Breakspear, 'Detecting depression: A comparison between spontaneous and read speech,' in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2013, pp. 7542-7546.

[2] N. Cummins, S. Scherer, J. Krajewski, S. Schnieder, J. Epps, and T. F. Quatieri, 'A review of depression and suicide risk assessment using speech analysis,' *Speech Communication*, vol. 71, pp. 10-49, 2015.

[3] M. R. A. Morales, P. Stefanov, and F. Ringeval, 'Speech vs. text: A comparative study of multimodal depression detection,' in *Interspeech*, 2018, pp. 2484-2488.

[4] F. Ringeval et al., 'AVEC 2019 workshop and challenge: state-of-mind, detecting depression with AI, and cross-cultural affect recognition,' in *Proceedings of the 9th International on Audio/Visual Emotion Challenge and Workshop*, 2019, pp. 3-12.

[5] J. R. Williamson, T. F. Quatieri, B. S. Helfer, G. Ciccarelli, and N. Malyska, 'Vocal biomarkers of depression based on motor coordination,' in *Interspeech*, 2013, pp. 3384-3388.

[6] F. Eyben et al., 'The Geneva Minimalistic Acoustic Parameter Set (GeMAPS) for Voice Research and Affective Computing,' *IEEE Transactions on Affective Computing*, vol. 7, no. 2, pp. 190-202, 2016.

# Parts 28 & 29 — Final Summary & 2-Minute Viva Script

## Final Technical Viva Script (2-Minute Summary):

*"In this Capstone project, we developed a participant-independent, multimodal screening prototype for depression detection using the clinical DAIC-WOZ dataset. Our system processes semi-structured interview recordings by extracting three complementary information streams: 154-dimensional acoustic features (openSMILE eGeMAPS, MFCCs, and prosodic statistics), 768-dimensional deep speech embeddings from frozen Wav2Vec 2.0, and 768-dimensional participant-only contextual linguistic embeddings from frozen BERT.*

*During our forensic investigation, we discovered that naive deep recurrent architectures (such as an 847k-parameter BiLSTM) severely overfit the small 116-session training cohort, collapsing into the 84% majority non-depressed class. To overcome this without data leakage, we implemented three key methodological corrections: (1) an automated transcript filter that strips virtual interviewer speech to prevent shortcut learning, (2) a 5-fold Stratified Group Split ensuring strict participant isolation and minority class representation across all folds, and (3) a regularized Compact Tri-Modal Multimodal MLP (225,413 parameters) trained with inverse-frequency class-weighted cross-entropy and validation Macro-F1 checkpoint selection.*

*On the held-out test split, our final CompactTriModalMLP achieved **83.78% accuracy, 42.36% Macro-F1, 48.92% Balanced Accuracy, 54.55% Binary F1, and 3.07 PHQ-8 MAE** (with 50.00% severe recall, 1/2 severe cases). When evaluated on 9 completely untouched unseen clinical sessions, the system attained **66.67% accuracy (6/9 correct)**, correctly identifying 4/4 Minimal/Mild cases, 1/2 Moderate cases (PID 409 at 71.6% confidence), and 1/3 Severe cases (PID 321 at 87.7% confidence). This provides preliminary evidence that compact multimodal fusion can improve participant-level generalization on this small, highly imbalanced evaluation set."*